From a single number to a landscape: topology-resolved quantum volume

Quantum Science and Technologies

Goal of the presentation

We study how Quantum Volume depends on *where* a circuit is placed on a chip, not only on the chip as a whole.

Structure

1. NISQ computers,
2. Noisy simulators,
3. Quantum volume,
4. Research demo

NISQ computers

- NISQ devices have useful scale, but are limited by noise and lack fault tolerance.
- Performance depends on qubit quality, connectivity, and compilation overhead.
- Therefore, system-level benchmarks are needed in addition to gate-level metrics.

Takeaway

In the NISQ era, *where* a circuit is placed on a chip can matter almost as much as *what* circuit is run.

Noisy simulators

- A noisy simulator reproduces hardware-relevant errors using a calibrated noise model.
- It enables controlled comparison of many circuit placements under identical conditions.
- In this project, calibration data and the coupling graph define the simulated device model.

Why simulation?

It is much easier to scan hundreds of connected subsets reproducibly on a simulator than on a live device.

All experiments require a physical$_q$*ubitsparameterasinputthatspecifieswhichphysicalqubitorqubitsthe*

Quantum volume (1)

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Heavy set of outputs

Let $\mathbf{Q}$ be a random circuit drawn from a suitable ensemble acting on n qubits. The quantum state after executing the circuit is denoted $|\psi\rangle$. Each possible output state $x \in \{0,1\}^n$ is measured with probability $|\langle x|\psi\rangle|^2$. The set of output states with a probability greater than the median constitutes the heavy set of outputs associated with the quantum circuit $\mathbf{Q}$.

Quantum volume (2)

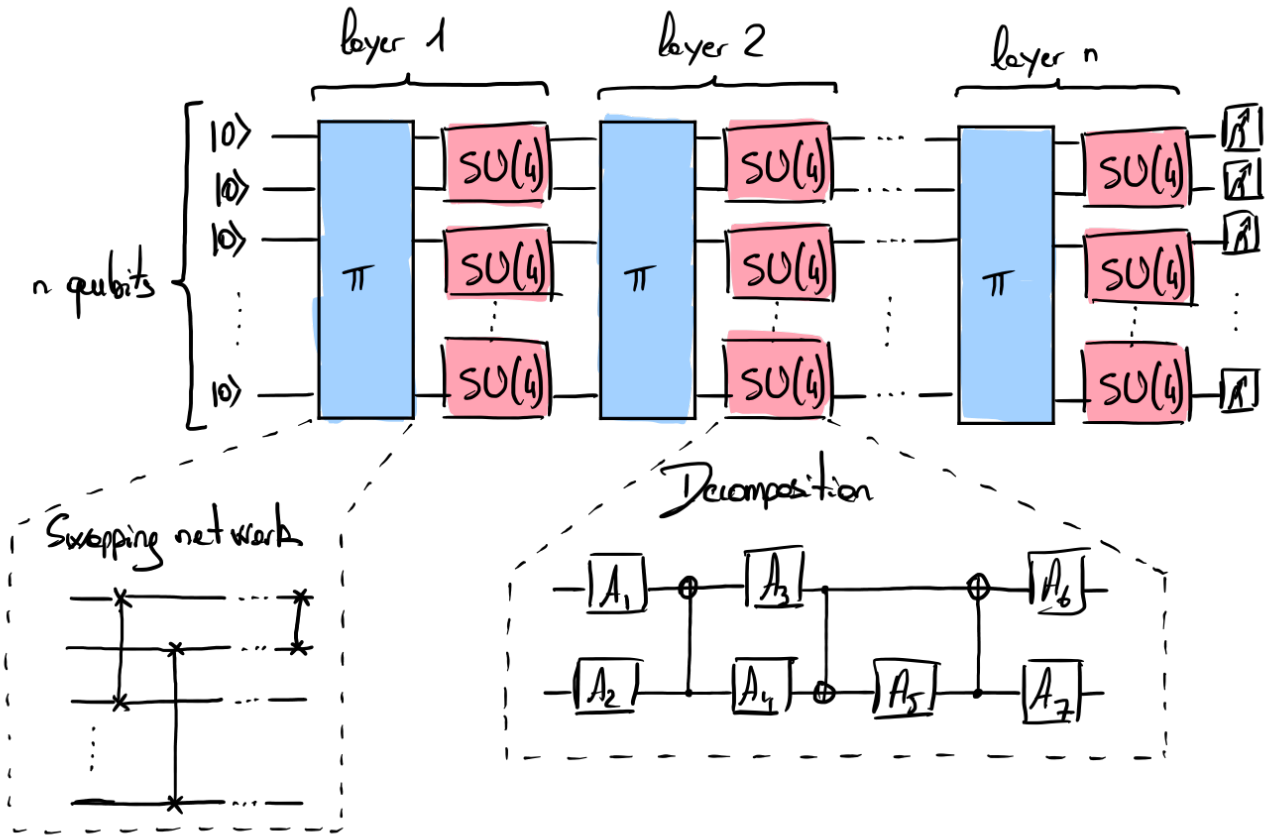

Figure: The QV circuit. Source: `https://quantumbenchmarkzoo.org/content/system-level-benchmark/others/quantum-volume`

Quantum volume (3)

If the quantum computer samples a distribution from the **heavy set of outputs** (simulated classically) with probability $> 2/3$, it validates the corresponding quantum volume score of 2^n. The confidence interval for this evaluation is set to 2σ (97.7%).

Problem

Quantum volume assesses only a limited subset of best-quality qubits within a quantum processor and does not provide a comprehensive measure of the overall fidelity or performance of quantum operations across the entire chip.

Simulator

First, if you want to model what the transpiler would do on the real device globally, then the full coupling map is more faithful. The transpiler uses the coupling map as a routing/optimization constraint, and transpilation is a multi-stage rewrite/optimization process. Giving it the full device graph versus only the induced subgraph can change the pass behavior and the final circuit, even if the final circuit still only touches the same qubits.

Research methodology

We benchmarked QV on the IBM Marrakesh device using calibration data and the coupling graph from Qiskit.

1. Construct a noisy quantum simulator from the extracted data
2. For each size $n \leq 6$:
    1. Enumerate all connected subgraphs of size n
    2. Run QV circuits on each subgraph
    3. Compute heavy output probability and confidence

Results (1)

Unoptimized circuits

n	Count	$\langle HOP \rangle$	$\sigma(HOP)$	Median	Min	Max	$\langle SE \rangle$	Lower
3	244	0.664	0.063	0.688	0.438	0.712	0.047	0.572
4	354	0.732	0.083	0.766	0.503	0.817	0.043	0.647
5	566	0.733	0.085	0.769	0.498	0.825	0.043	0.648
6	926	0.748	0.085	0.779	0.491	0.832	0.042	0.665

Optimized circuits

n	Count	$\langle HOP \rangle$	$\sigma(HOP)$	Median	Min	Max	$\langle SE \rangle$	Lower
3	244	0.667	0.063	0.692	0.433	0.715	0.047	0.575
4	354	0.742	0.084	0.776	0.495	0.821	0.043	0.658
5	566	0.741	0.085	0.778	0.502	0.828	0.043	0.657
6	926	0.701	0.087	0.738	0.492	0.811	0.045	0.613

Results (2)

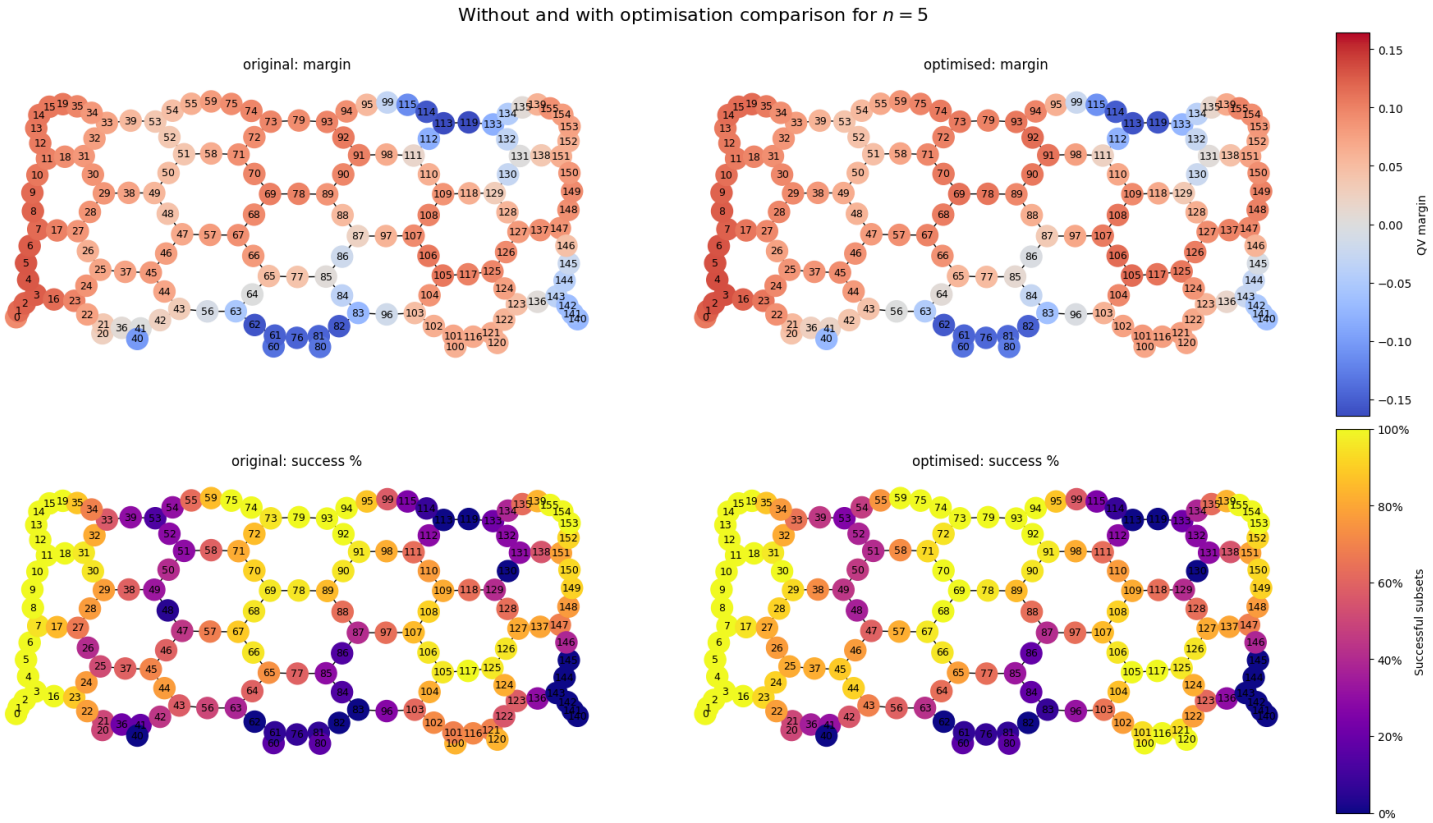

Figure: Without and with optimisation comparison for $n = 6$

Discussion

For QV, each layer is built from random SU(4) two-qubit blocks after a random permutation, and for odd widths one qubit is idle each layer. On a connected 3-qubit path like (a, b, c), one of the possible two-qubit interactions is between the two endpoints. That interaction is not native, so with trivial layout and basic routing it tends to introduce SWAP overheadresulting in extra routing cost that can dominate the whole layer. 3Q layers contain only one gate, so routing dominates. 4Q layers contain two gates, allowing routing overhead to be shared.

Results (3)

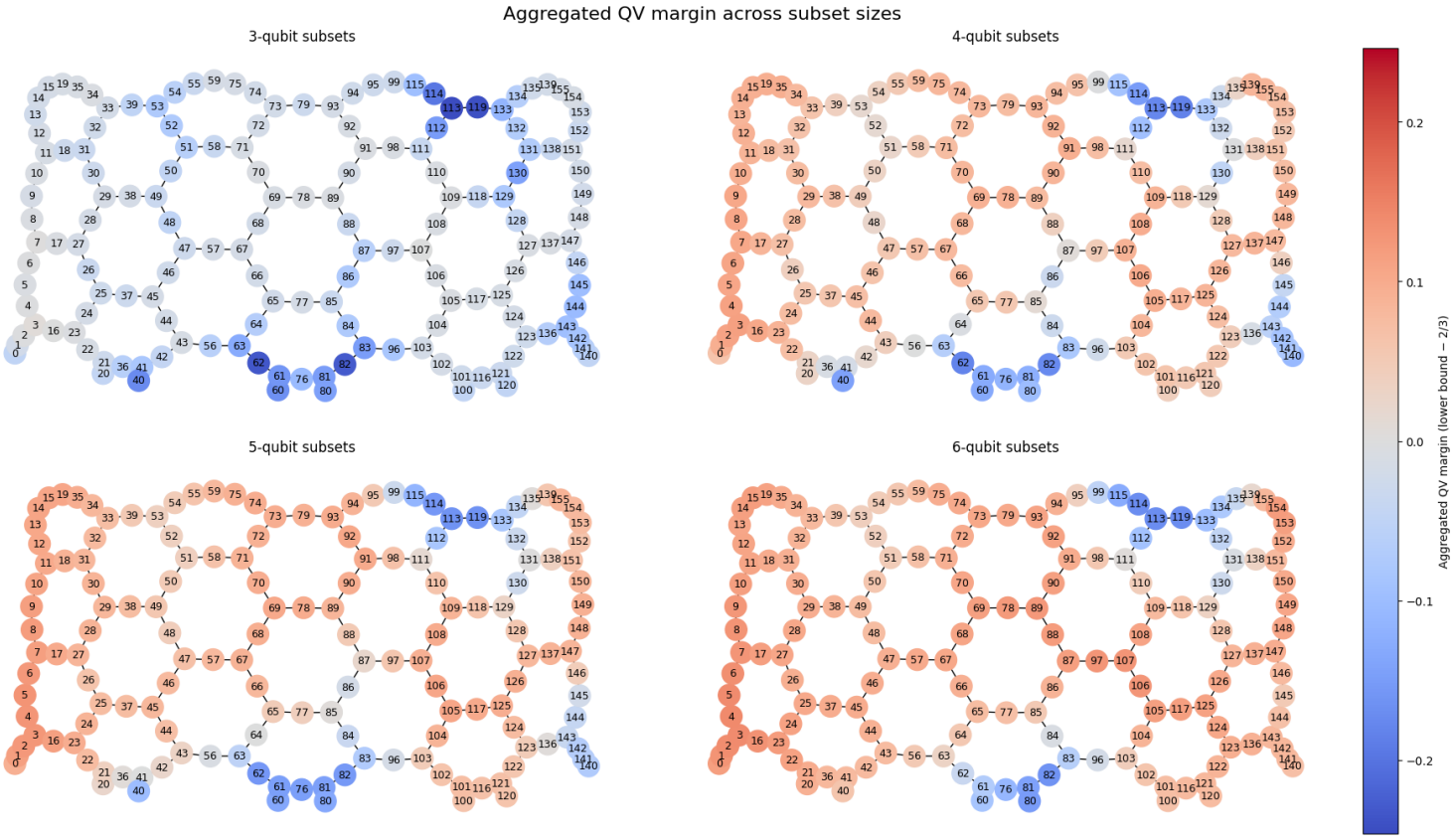

Figure: Aggregated QV margin across subset sizes

Results (4)

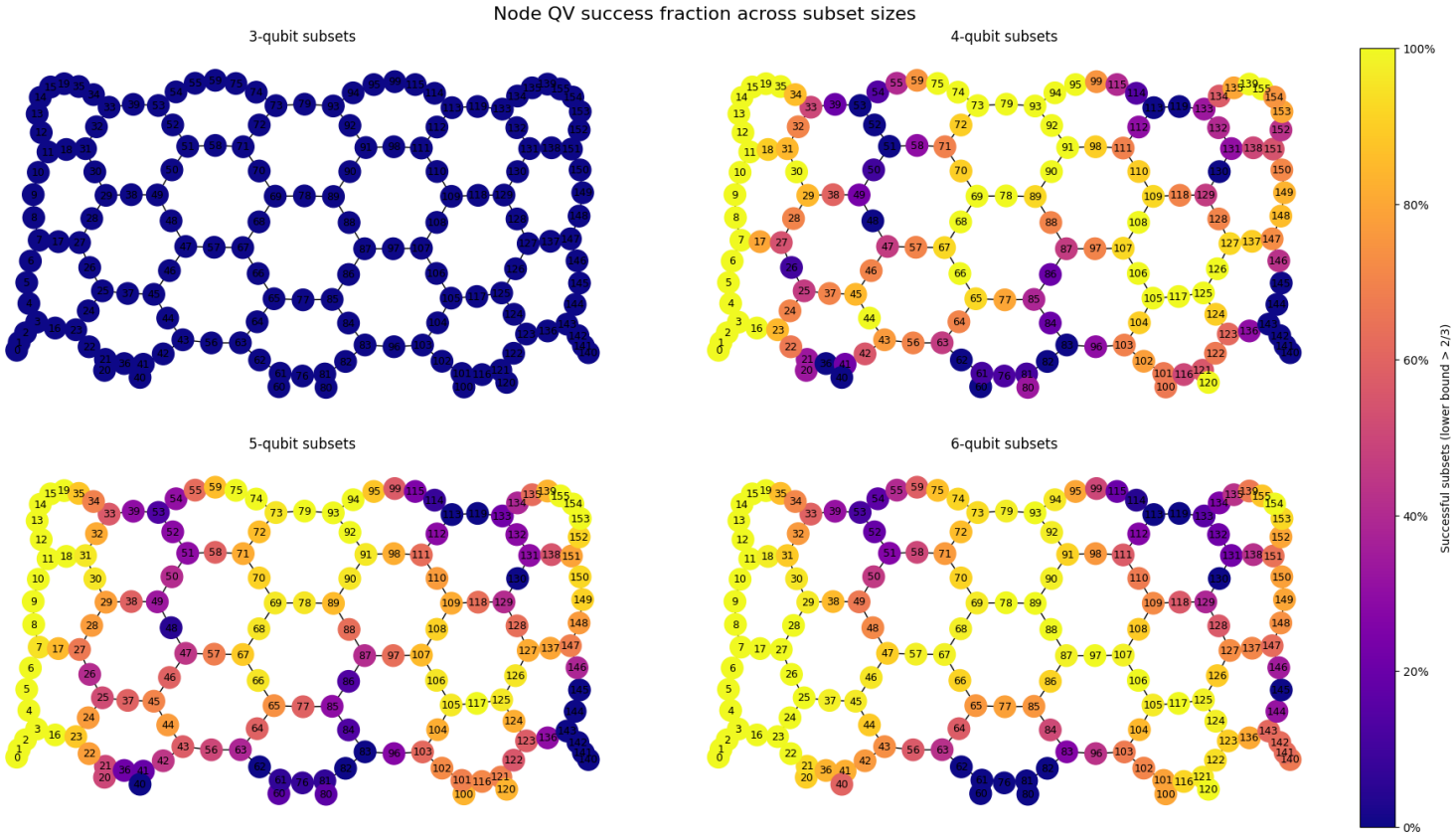

Figure: Success fraction of QV circuits across subset sizes

Results (5)

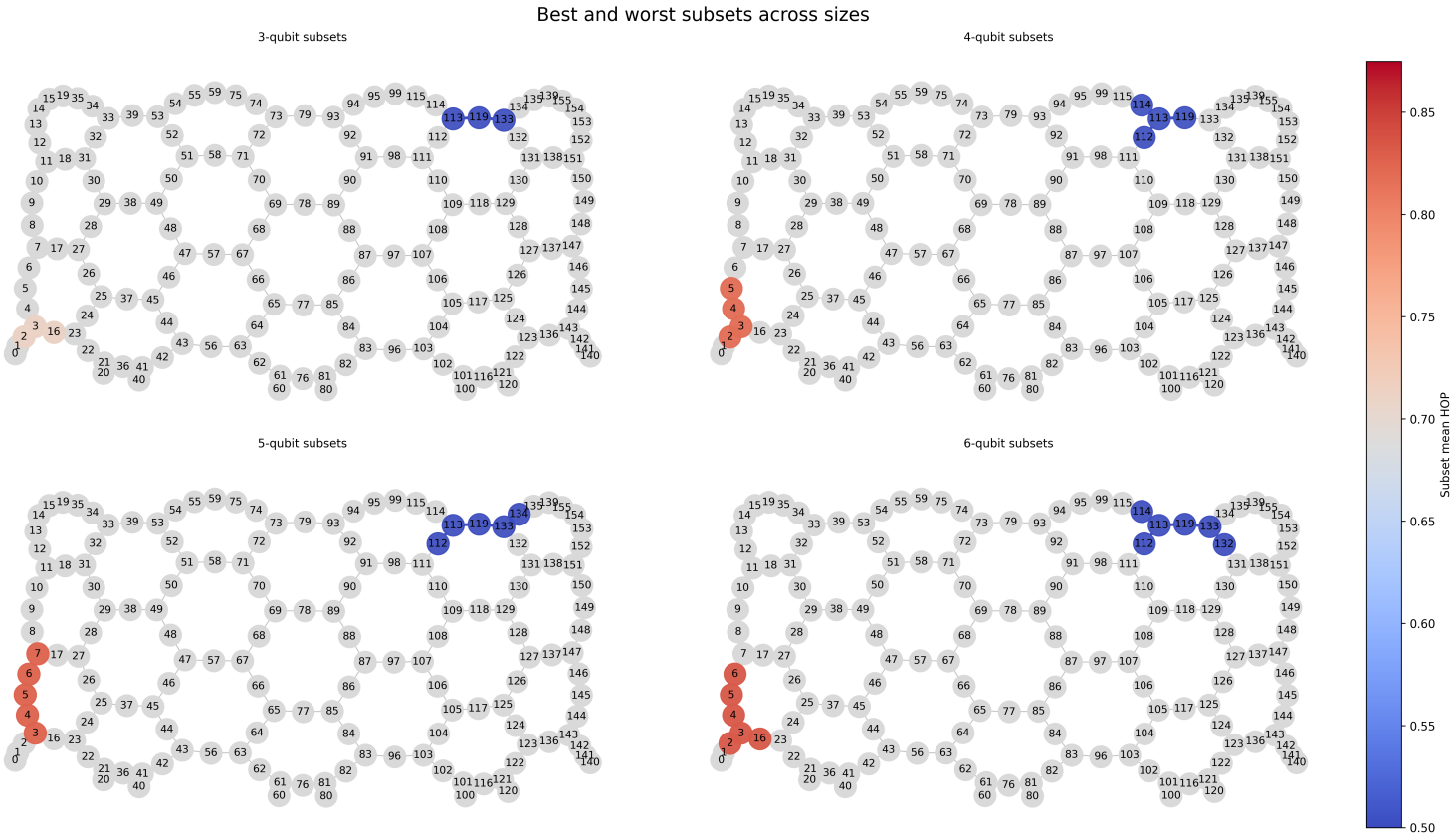

Figure: Best and worst performing subsets across subset sizes

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